(a) Prefix reuse

Prompt You are maintaining a Python repository. Diagnose the timeout failure below.

```
deadline = monotonic() + timeout
connection = await client.acquire()
```

Action	GPU KV		LMCache DRAM
1. Build Original prompt	534 computed Encode 4 closed blocks	Copy →	4 original-prefix blocks
2. Append Add a regression test	512 reused + 144 computed Encode 1 new block	Copy →	+1 new suffix block
3. Edit early "Python" → "TypeScript"	0 reused + 534 computed Encode 4 new blocks	Copy →	+4 edited-prefix blocks
4. Restore Replay original prompt	512 restored + 22 computed After GPU cache reset	← Import	4 original-prefix blocks

Restore response

"…it is not enforced during the `client.acquire()` call."
TTFT **47.3 ms**

KV Inspector (Build)

18.00 MiB → **12.73 MiB**
0 BF16 mismatches

(b) GPU capacity

54 GiB KV / GPU		Raw	Pool mode
KV tokens per GPU	↑	**1.18M**	**1.59M**
Waiting requests (mean)	↓	**1.5**	**0.4**
Engine TTFT (s)	↓	**5.16**	**1.94**

(c) History retention

400 GiB DRAM		Raw	Tier mode
Cached tokens (GPU + ext.)	↑	**55%**	**91%**
Stored / raw bytes	↓	**100%**	**71.6%**
Engine TTFT (s)	↓	**6.76**	**2.34**